

FSAE-A AV ACCUMULATOR INSPECTION VIDEO

2025

Revision Table

Date	Revision	Comment	Initial
20250819	1.0	Draft	N.O
20250916	1.1	Added submission link	N.O

Submission & Dates

4 weeks prior to competition

Note: *This is an optional task for teams for the 2025 competition year.*

Teams have the option to submit a video demonstrating their accumulator inspection as part of the pre-event scrutineering process. A Google Form has been provided for all relevant submissions, including a section specifically for uploading the link to your accumulator inspection video. While this submission is optional, it is strongly encouraged to help streamline the on-site inspection process. The form can be accessed at:

<https://forms.gle/NeMtX7eQbWfbWKms5>.

Purpose

- To help FSAE technical inspectors become familiar with your accumulator design before the event and to optimise and minimise the inspection time during technical inspection at competition.
- Provide teams the opportunity to be prepared for accumulator inspection
- Aims to streamline on-site inspection, enabling early detection and correction of issues at your workshop rather than under competition pressure—and potentially secure a lower Tech Order Number.

Key Elements to Show in Your Video

1. **Inspection Points:** Follow the accumulator inspection sheet step-by-step—for example, show thermal strip placements and inspection, segment interconnectors, visual evidence of insulation, and assembly details.
2. **Inspection Holes / Access Zones:** Demonstrate clear access for inspectors to check internal components without disassembly.
3. **Thermal Strip Locations:** Clearly point out where temperature sensors are placed—on negative cell terminals or busbars with proper coverage (min. 30% of cells)
4. **Segment Interconnectors & Assembly Process:** Show how segments are connected and secured, ensuring they're tool-free to disconnect and using positive locking where required.
5. **Housing Insulation & HV Routing:** Highlight how the housing keeps high voltage isolated (no conductive breaches), and how warning labels or barriers are installed
6. **HV Connector & Charger Behavior:** Illustrate connector shutdown behavior—AIRs opening, voltage dropping below 60 V, and IMD functionality with proper ground paths
7. **Use of Safety Equipment:** Demonstrate use of safety equipment and PPE as appropriate to your design when working on your accumulator.
8. **Camera Movement:** Pan slowly around each component so inspectors can visually absorb the design—not just still shots



Video Brief Script Overview

Opening Title Card

- Team Name, Year, Car Number
- “Accumulator Inspection – Pre-Competition Video Submission FSAE Year 20##”

1. Introduction (~15 sec)

- Voiceover: “This video presents our accumulator for the (year) FSAE Electric event to assist inspectors’ pre-competition review.”

2. Inspection Sequence (~4 min)

- Start with the **closed housing**.
- Show **inspection/access holes**.
- Point out **thermal sensors** on cells/busbars—and specify they cover $\geq 30\%$ of cells.
- Show **segment interconnectors**, insulation, and how segments disconnect tool-free.
- Highlight **housing insulation** and deliberate HV/LV separation distance (measure visible).
- Demonstrate HV connector: attach charger, voiceover: “Charger connected—starting charge now.” Then press shutdown:
 - Show AIRs opening
 - Voltage dropping $< 60\text{ V}$
 - Show IMD disconnect path with ground wiring

Show **positive locking, ventilation, labels, and fuse/relay separations** if visible.

3. Safety & Compliance (~4 mins)

- Discuss work practices and risk mitigation strategies
- Illustrate correct use of tools and PPE
- Confirm team member training



4. Summary

- Voiceover: “This walkthrough follows the accumulator inspection sheet. All required design elements—including thermal monitoring, segment interlocks, safety shutdown, insulation, and labeling—are clearly shown.”

Final Tips

- **Print the accumulator inspection sheet** and follow it in order—you’ll ensure nothing is missed.
- **Prefer simplicity (“KISS” principle)**—a clear, well-organised video is better than over-complicated visuals.